

Lab 1 Test PCBs and Setup ECEN 299 Library and Project

4/16/2025

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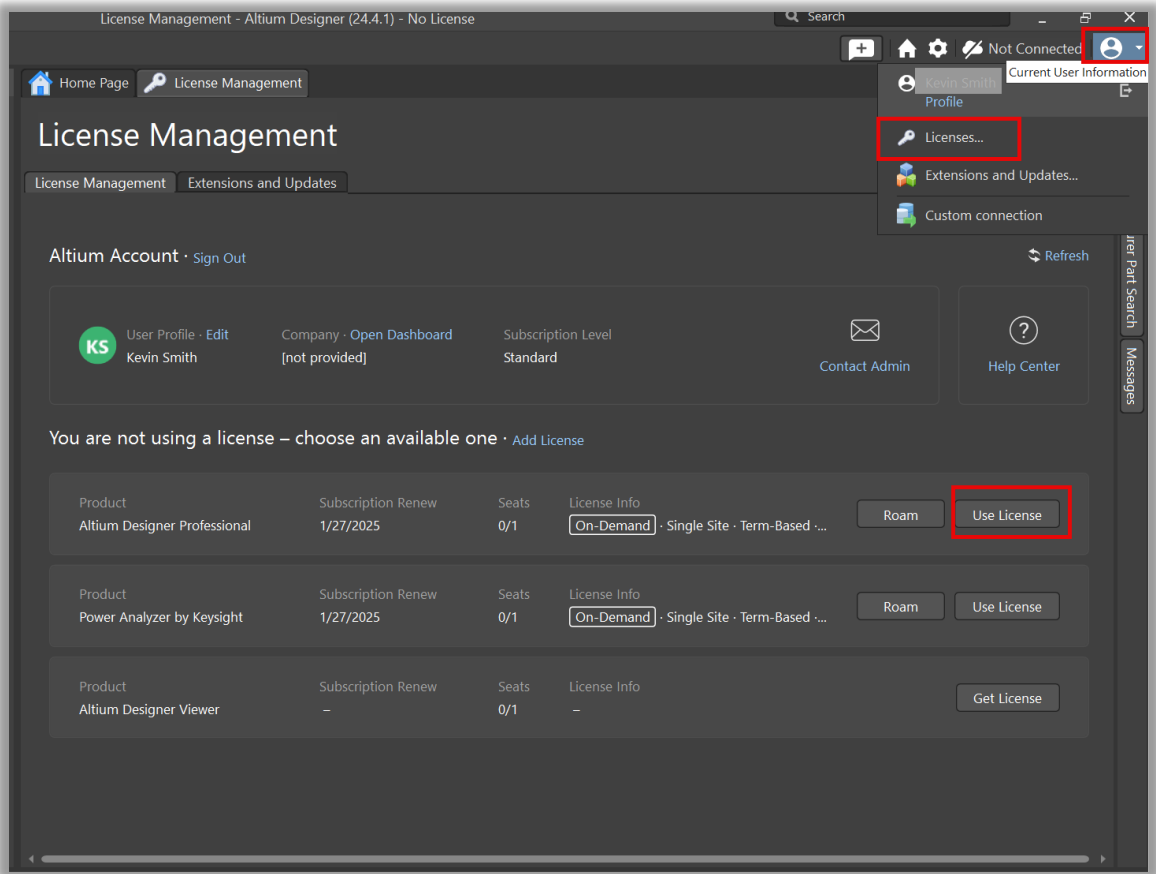
Objectives

The objective of this laboratory is to provide you with a general overview of the series of labs and assessments you will be working on during the rest of the semester. You will receive circuit boards from students who completed the labs in previous semesters, and you will verify that they work. After that, you will set up your ECEN 299 library and create a PCB project for your line buffer.

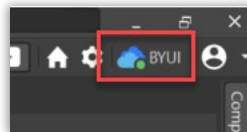
Lab 1 Part A: Setting Up Altium

Open Altium Designer and Use Your Student License

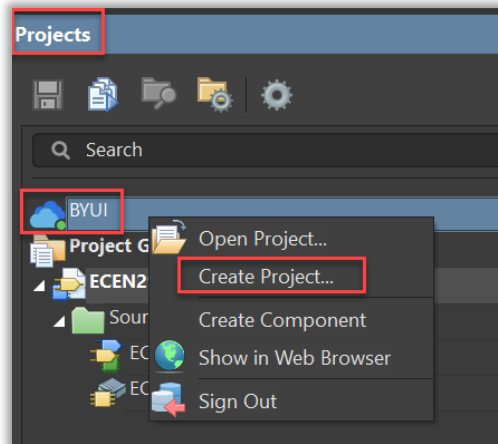
1. Select Windows Open > Altium Designer.
2. Sign In.
3. Select Current User Information > Licenses > Use License.



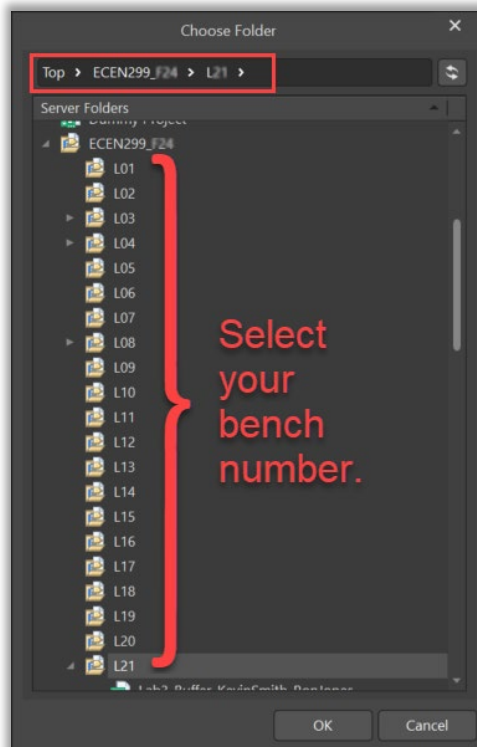
4. Close Licenses window.
 - a. Right click on the license management tab and select close license management.
5. Select BYUI as your active server.



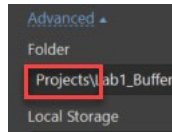
6. Select View > Panels > Projects.
7. Right click BYUI.
8. Select Create Project



9. Enter Lab1_Buffer_yourName_labPartnerName for the Project Name where xx is your lab bench number, yourName is your first and last name, and labPartnerName is the first and last name of your lab partner.
10. Select Version Control, if not already selected.
11. Select Constraint Management, if not already selected.
12. Select Advanced.
13. Select the ellipsis (...) for the Folder.
 This is the folder on the BYUI Workspace where the project will be stored.
14. Expand ECEN299_syr (where syr is the semester and year, e.g., W25)
15. Select the Lxx folder you are assigned.

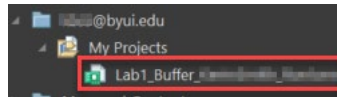


Do not use the default folder. The default folder “Projects\xxx”



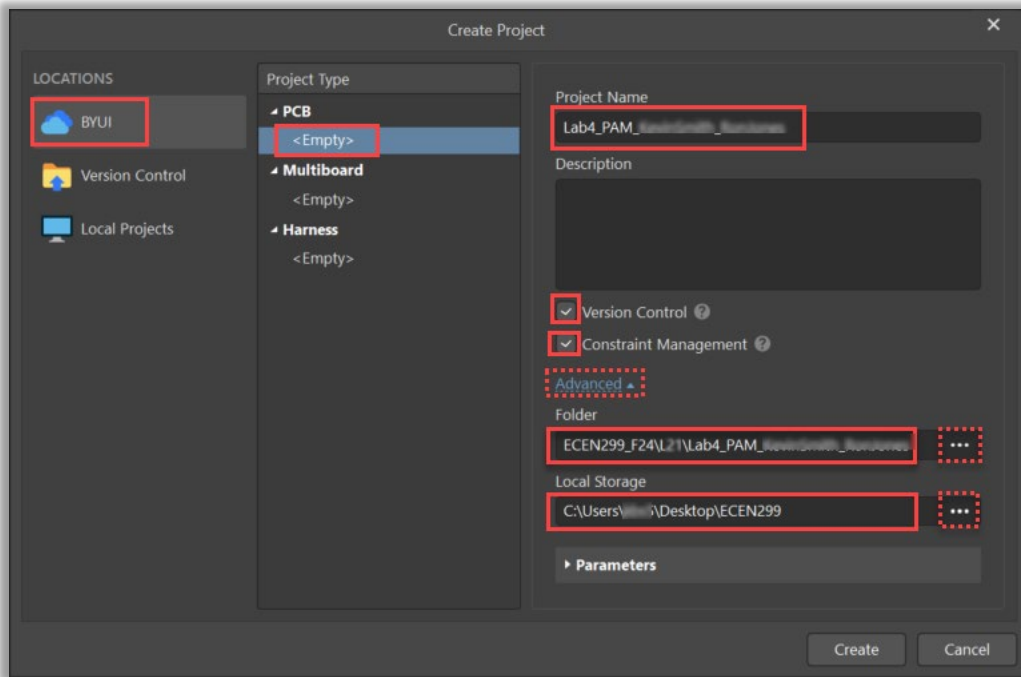
will store the project in a directory named BYUI > youremail@byui.edu > Projects. This folder will be created if it does not already exist. If the default folder is used by mistake, then

1. Open the Explorer panel.



2. Right click on the project and select Move Folder.
3. Browse to and select the correct folder where the project should be saved.
4. Delete the My Projects and youremail@byui.edu folders.

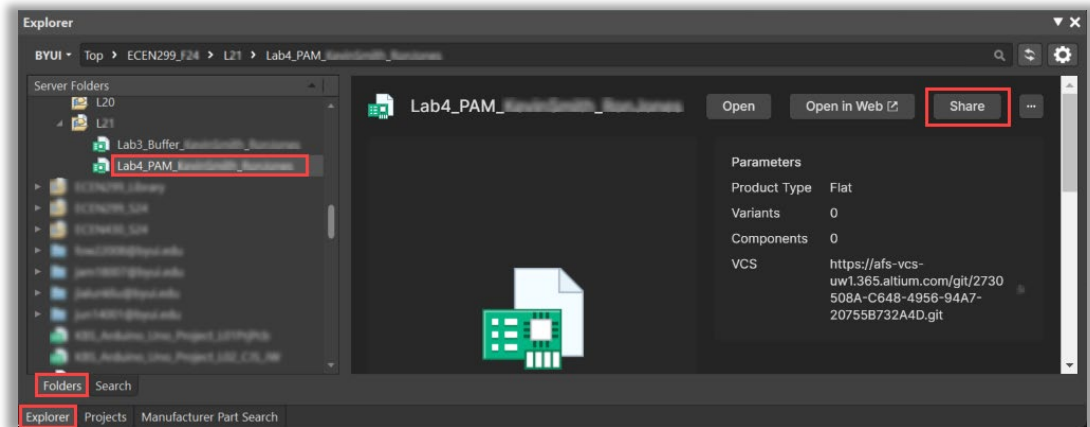
16. Select the ellipsis (...) for the Local Storage. This specifies the folder where a working copy of the project will be stored locally on the computer.
17. Browse to %UserProfile%\Desktop\ECEN299.
18. Click Select Folder to close Browse for project location.
19. Click OK to close Choose Folder.



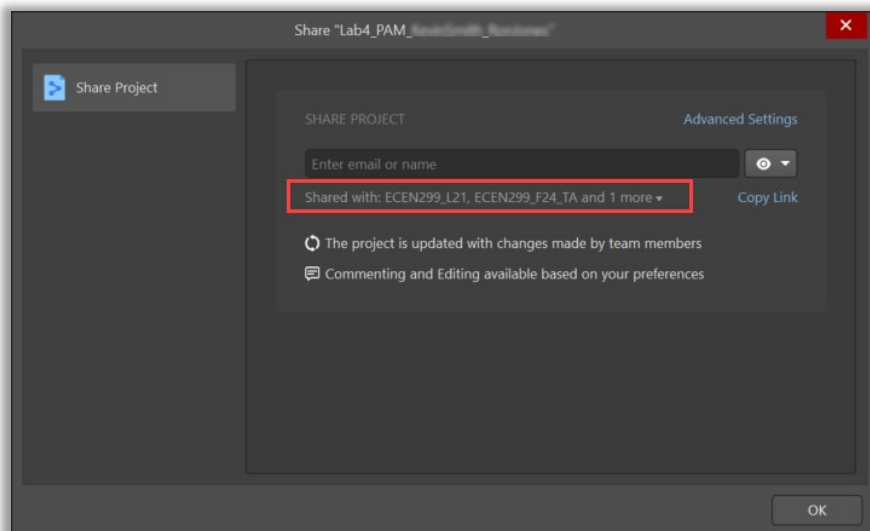
20. Select Create.

Edit the Shared.with Project Options

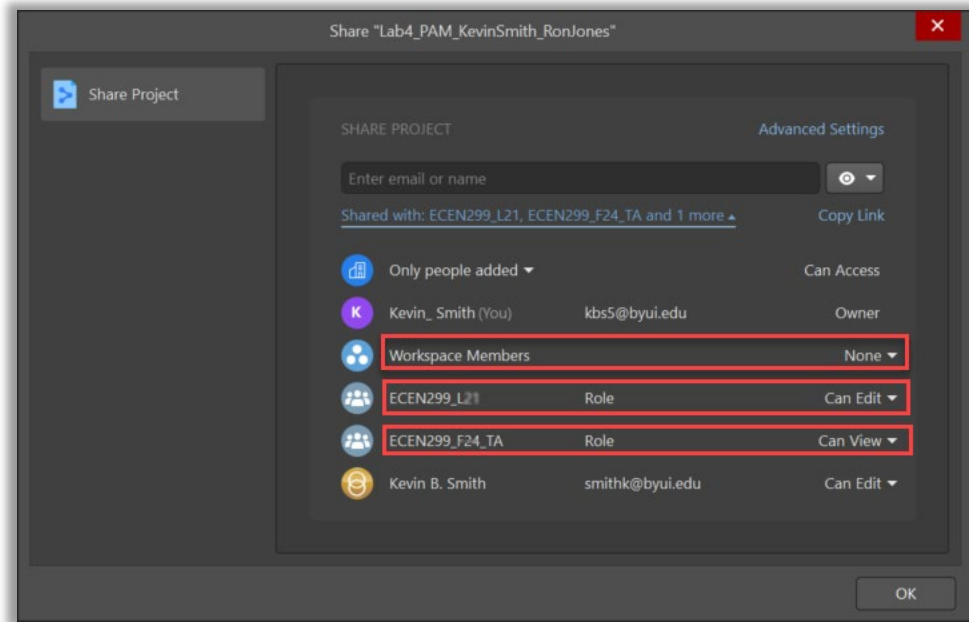
1. Select Panels > Explorer.
2. Select Folders tab at the bottom.
3. Expand the ECEN299_W25 > Lxx folder where xx is your lab bench number.
4. Select Lab1_Buffer_Lxx_yourName_labPartnerName.



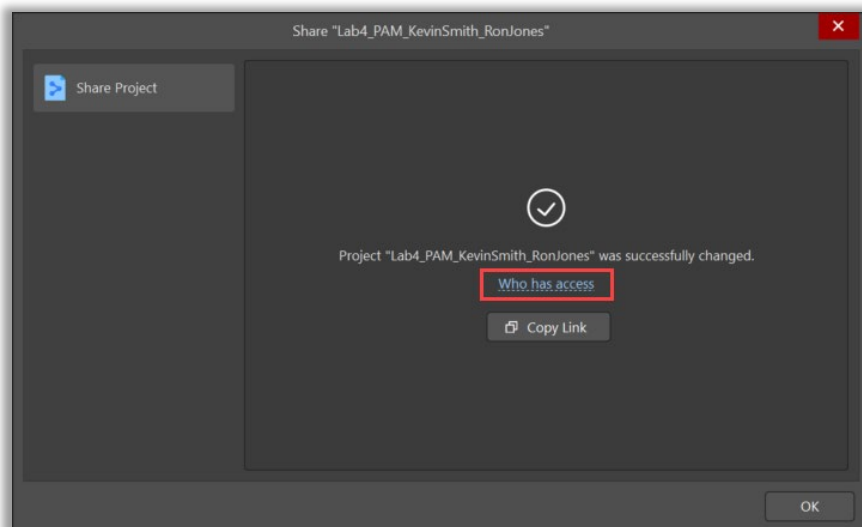
5. Select Share. You may have to expand the Explorer window to see the Share button.
6. Select Shared with: ECEN299... to see an expanded list.



The Share list should look like this when you are done:



7. Use the pulldown menu to change what a given role Can.Access. Make it match what is shown above.
8. The ECEN299_F24_TA will be the same but the semester your current semester. (e.g., ECEN299_W25_TA)
9. Select Save to save a change.
10. Select Who has access to return to the Share list.



Remove ECEN299_xxx_All where xxx is your current semester (e.g, W25)

1. Select Remove for ECEN299_F24_All.
This will prevent other teams from having access to your project.
2. Select Save.

Add ECEN299_Lxx, if it is not shown.

If ECEN299_Lxx is not shown:

3. Enter ECEN299_Lxx in the Enter email or name field.
4. Select ECEN299_Lxx (role).
This is a role that is assigned to you and your lab partner.
5. Select the eye icon and select Can Edit.
6. Select Share.

Add ECEN299_F24_TA = Can View, if it is not shown.

If ECEN299_F24_TA does not exist:

1. Enter ECEN299_F24_TA in the Enter email or name field.
2. Select ECEN299_F24_TA (role).
3. Select the eye icon and select Can View. (if not already selected)
4. Select Share.

Remove Workspace Members from accessing:

If the access for Workspace Members is not None:

7. Select Remove for Workspace Members.
8. Select Save.

11. Select OK to close Share "Lab1_Buffer_Lxx...".

This will only give you and your lab partner access to edit the project, and it will give the TAs access to view your project.

Add Schematic Sheet

1. Select View > Panels > Projects.
2. Right click on Lab1_Buffer_yourName_labPartnerName.PrjPcb.
- 9j Select Add New to Project > Schematic document. (If.SCH.list.shows.up.you.can.drag.it.to.the.side.of.the.screen.or.park.it.to.the.side.of.the.main.windowj)
4. Right click on Sheet1.SchDoc.
5. Select Save.
The default directory is the project directory: Desktop\ECEN299\Lab1_Buffer.
6. Enter File name: Lab1_Buffer_Sheet1.SchDoc.
7. Select Save.

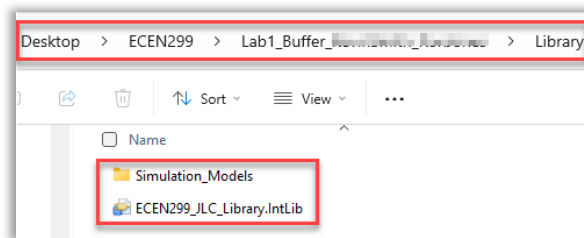
Add a PCB Layout Document

1. Select View > Panels > Projects, if the Projects panel is not already open.
2. Right click on Lab1_Buffer_yourName_labPartnerName.PrjPcb.

3. Select Add New to Project > PCB.
PCB1.PcbDoc is created.
4. Right click on PCB1.PcbDoc.
5. Select Save.
The default directory is the project directory: Desktop\ECEN299\Lab1_Buffer.
6. Enter File name: Lab1_Buffer_PCB1.PcbDoc.
7. Select Save.

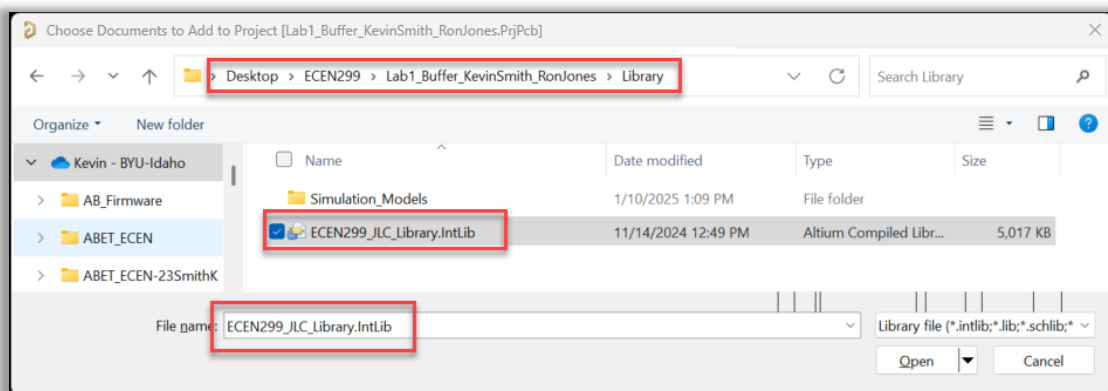
Copy the Integrated Library

1. From Canvas > Lab1, download ECEN299_JLC_Library.zip.
2. Extract this .zip file.
3. Select and copy the file ECEN299_JLC_Library.IntLib and the folder Simulation_Models,
4. Browse to %UserProfile%\Desktop\ECEN299\Lab1_Buffer_yourName_labPartnerName\
5. Create a folder named Library
6. Open the Library folder.
7. Paste the file and folder here.

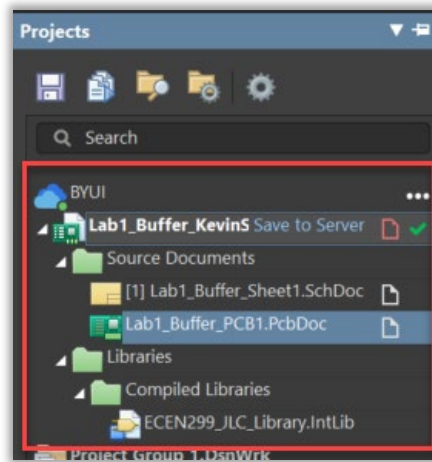


Add Library to the Project

1. Go back to Altium Designer.
2. Right click on Lab1_Buffer_Lxx_yourName_labPartnerName.PrjPcb.
3. Select Add Existing to Project.
4. Browse to
Desktop\ECEN299\Lab1_Buffer_Lxx_yourName_labPartnerName\Library.
5. Select file type: Library file (*.intlib, *.lib, *.schlib,...)
6. Select ECEN299_JLC_Library.IntLib.

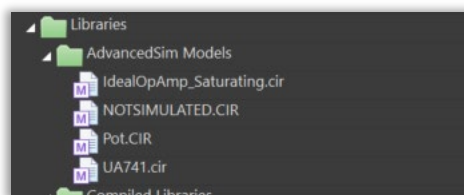


7. Select Open.
8. Verify that Projects panel looks like this.



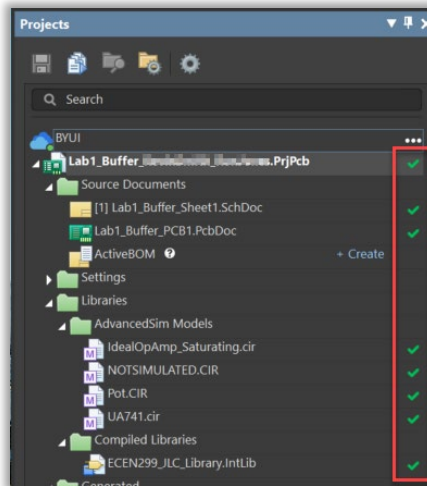
Add Simulation Files to the Project

1. Open the Projects panel, if it is not already open.
2. Right click on Lab1_Buffer_Lxx_yourName_labPartnerName.PrjPcb.
3. Select Add Existing to Project.
4. Browse to
Desktop\ECEN299\Lab1_Buffer_Lxx_yourName_labPartnerName\Library\Simulation_Models.
5. Select file type: All files (*.*).
6. Select all four .cir files.
7. Select Open.
8. Expand Libraries > AdvancedSim Models.
The models should now be added to the project.



Save the Project to the Server

1. Right click on Lab1_Buffer_Lxx_yourName_labPartnerName.PrjPcb.
2. Select Save to Server.
3. Select OK to Save Lab1_Buffer_....

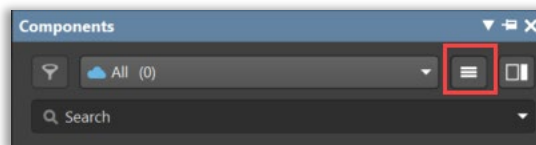


4. Verify that green checkmarks are displayed next to the project files. The check marks indicate that the files are saved to the server.

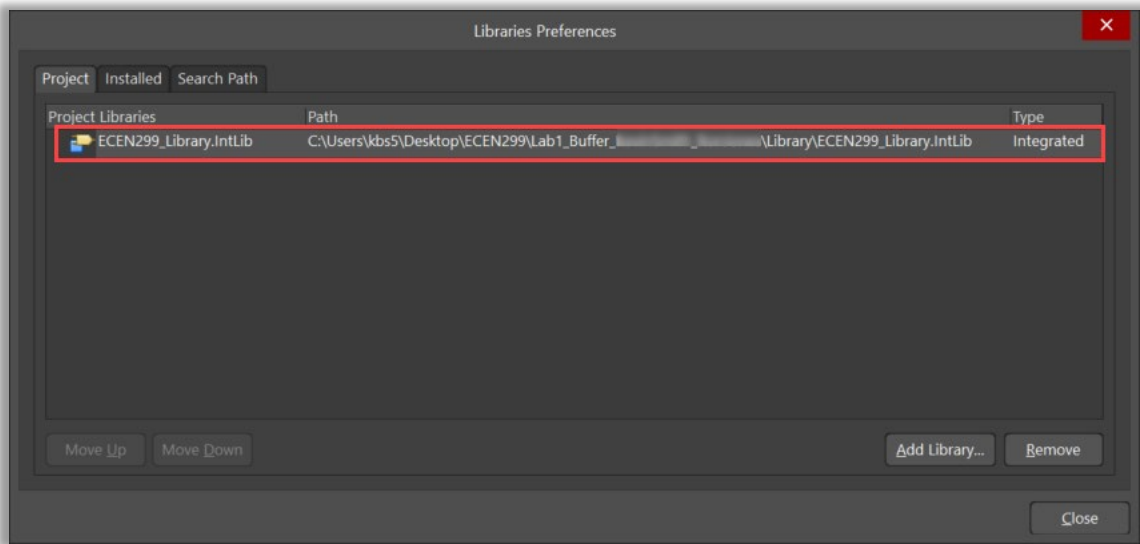
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Configure Altium Designer to reference the Integrated Library

1. Go back to Altium Designer.
2. Open the Components panel (Select View > Panels > Components).
3. Select the Operations icon.

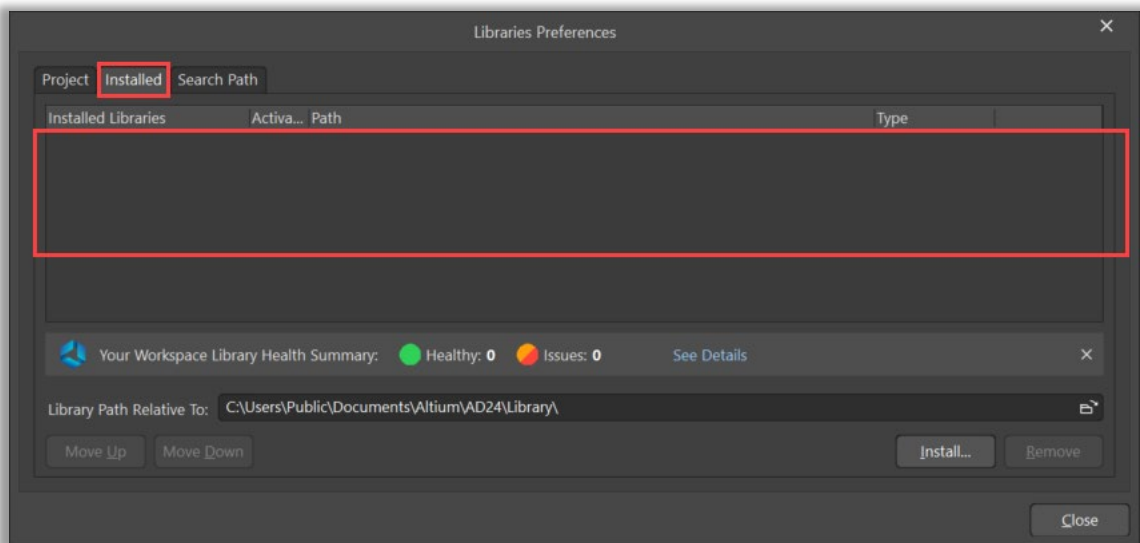


4. Select Library Preferences.
5. Select the Project tab, if not already selected.
6. Verify that the integrated library is the only one listed.



If there are others listed, remove them.

7. Select the Installed tab.
8. Verify that there are no Installed Libraries.

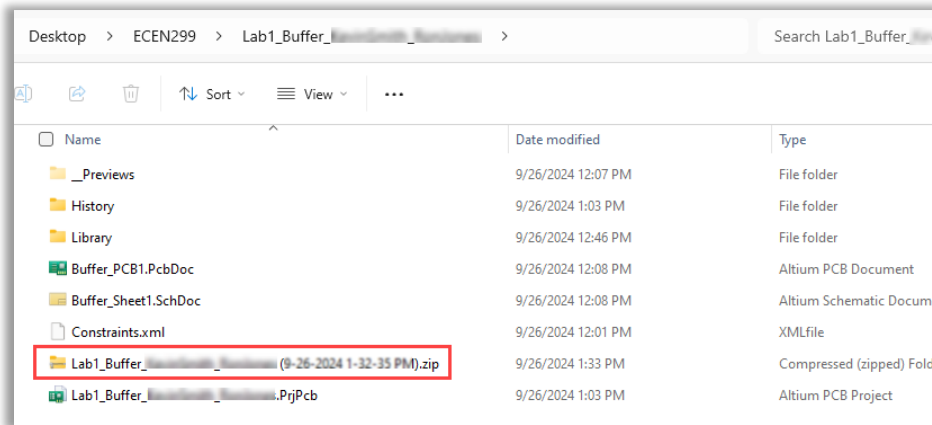


If there are any, Remove them.

9. Select Close to close Libraries Preferences.

Select File > Save All. Then create a Zipped archive of the project files

1. Right-click on the project file (Lab1_Buffer_yourName_labPartnerName.PrjPcb)
2. Select Project Packager (or select Project tab > Project Packager).
3. Select Next, Next, Next, and Finish to accept the defaults.
4. Right-click on the project.
5. Select Explore to see the zip file just created.



6. Upload this zip file to I-Learn.

Grading Rubric

Refer to the grading rubric on I-Learn for a summary of what we are looking for; it may also contain additional requirements or clarifications not included in these instructions.

Lab 1 Part B: Introduction to the Stereo Amplifier

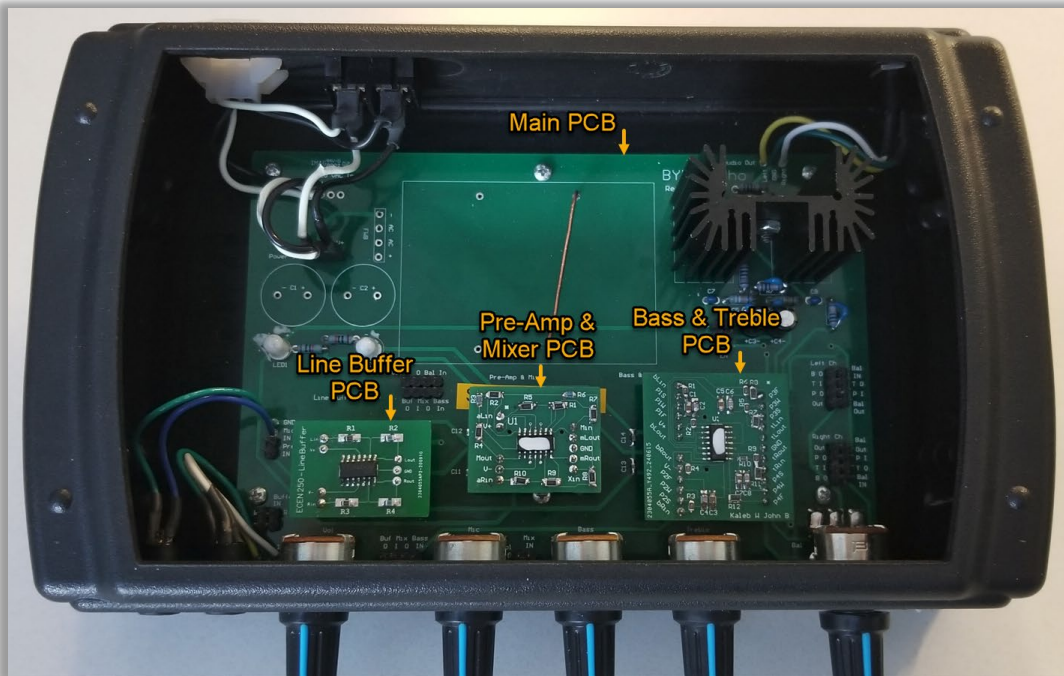
Principles to Be Studied

- High gain amplifier design
- Operational amplifier filter circuit design
- Voltage dividers and variable resistors
- Op-amp virtual short characteristics and input impedance
- Signal clipping
- High gain amplifier design

Overview

There are three Printed Circuit Boards (PCBs) that you will create in this class. The first is a line buffer, which prevents the amplifier from excessively loading the source of the audio input signals that are plugged into the 1/8" stereo jack. The second PCB is a preamplifier for the microphone and a mixer that combines the microphone signal with the audio input signals. The third PCB contains the bass and treble circuitry. These three PCBs (daughter boards) plug into the main PCB (motherboard). There are jumpers on the motherboard that allow it to be configured with different combinations of daughter boards. This enables the motherboard to be used with various configurations of daughter boards.





Procedures

Stereo Amplifier: Motherboard Jumpers and Daughterboard Header Pins

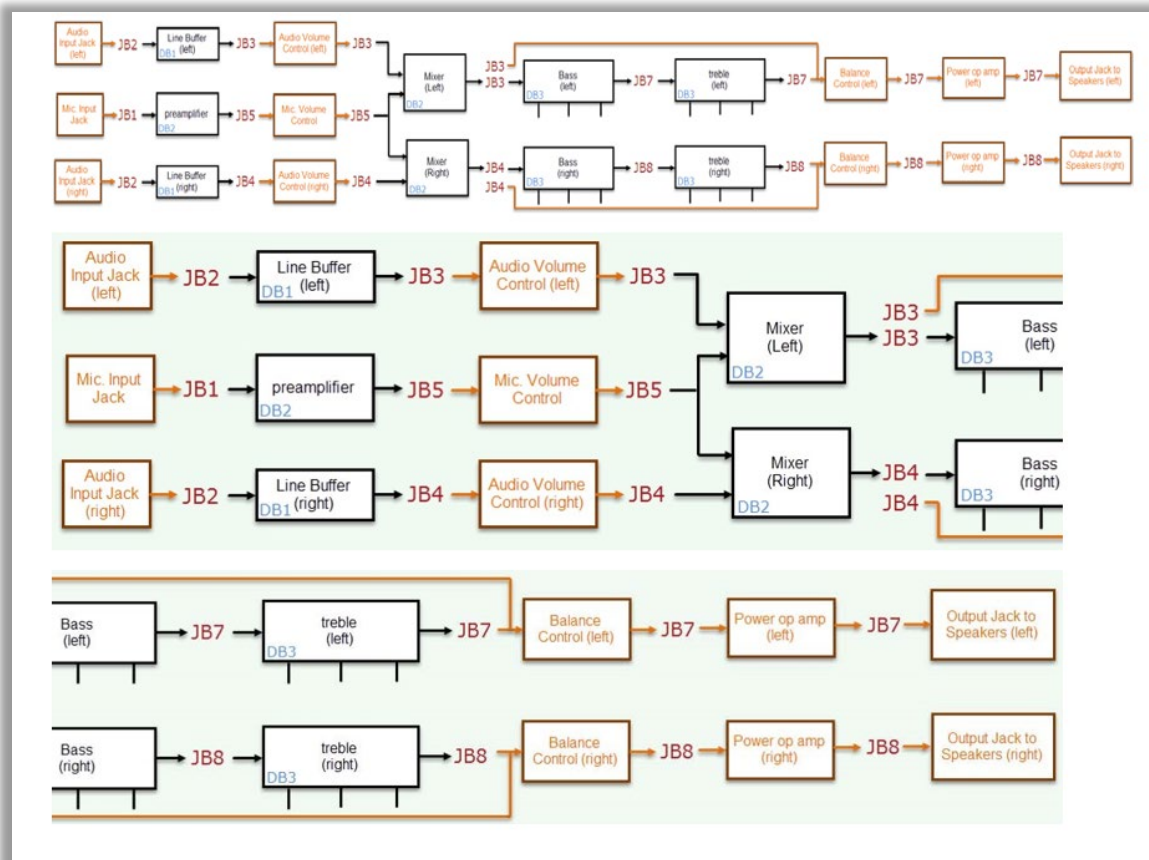


Figure.7.Logical.Flow.of.Stereo.Amplifier.System.showingjumper.connections.(JB-);

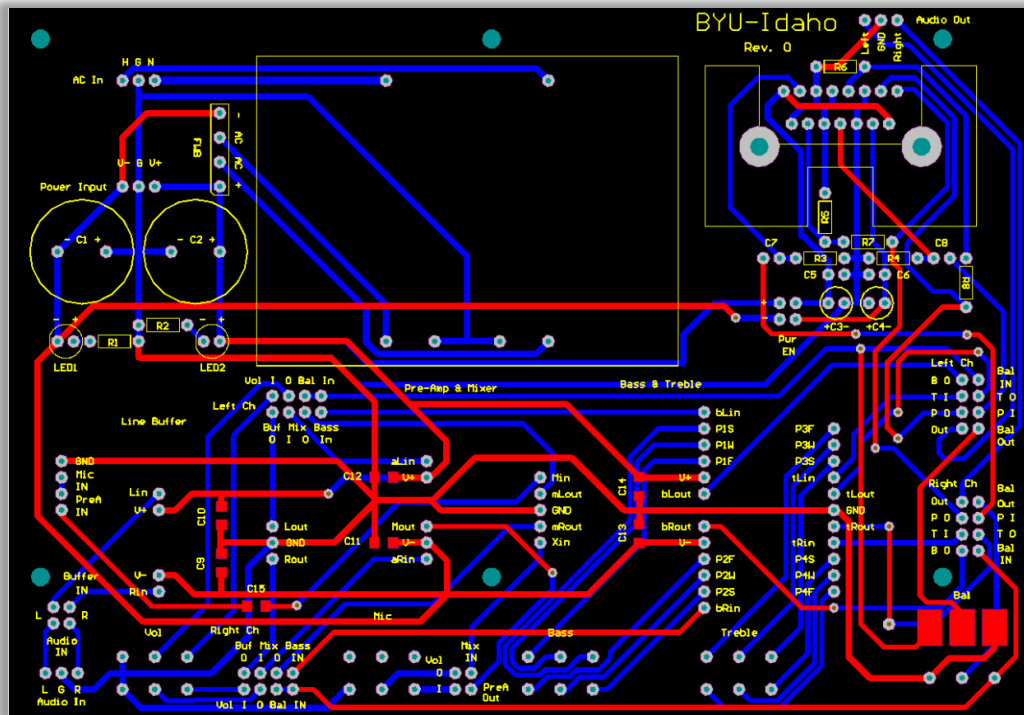


Figure.8.Layout.of.Stereo.Amplifier.Motherboard;

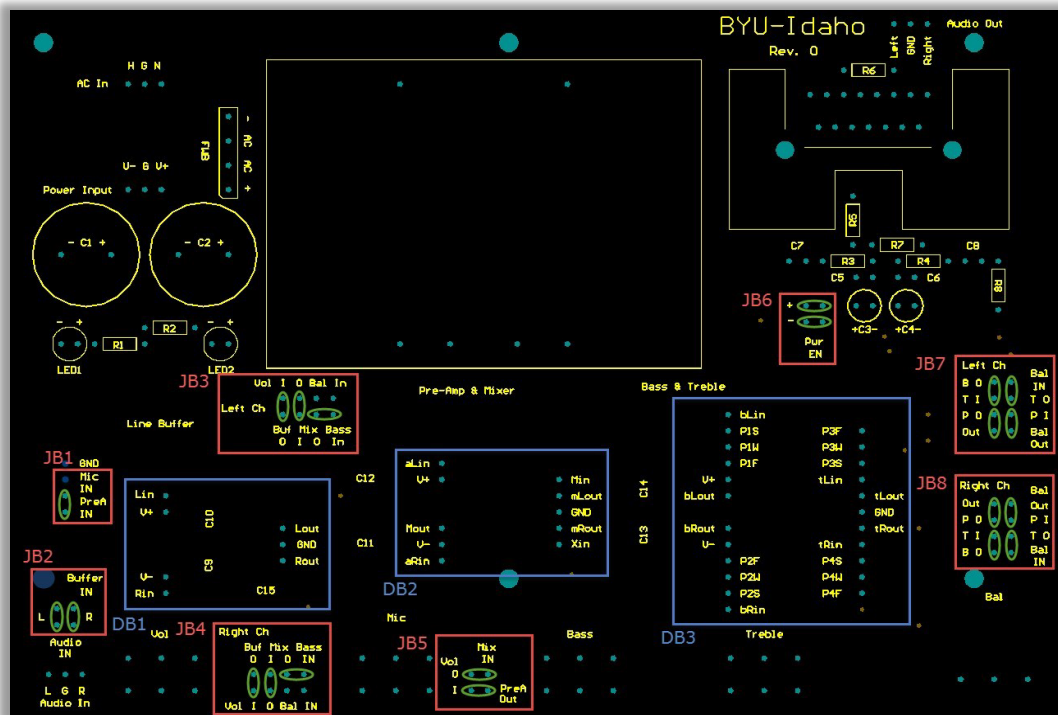


Figure.9.Stereo.Amplifier.Motherboard.with.Jumper.Blocks.(JB).and.Daughterboard.(DB).header.pins.highlighted;

Make sure you can identify the regions labelled JB1 to JB8 and DB1 to DB3 in Figure 3. The region JB1 to JB8 are for jumper connections and DB1 to DB3 are slots to insert the daughter PCB boards.

Note: JB6 shows jumpers to connect power to the power amplifier. Since we are using powered speakers, we will bypass the power amplifier, so these connections are not needed.

To understand the basic operation of the stereo amplifier, watch the following video.

Video: [Overall System Operation](#)

Verify Functionality of Complete System with all three PCBs

To verify overall system operation, connect the line buffer PBC, preamp & Mixer PCB, as well as the treble and bass PCB (with the power turned off) to the stereo amplifier by plugging the boards into the proper slots on the mother board. Set the jumpers to route the signal appropriately as shown in Figure 1. For reference, the complete schematic is shown in Figure 2.

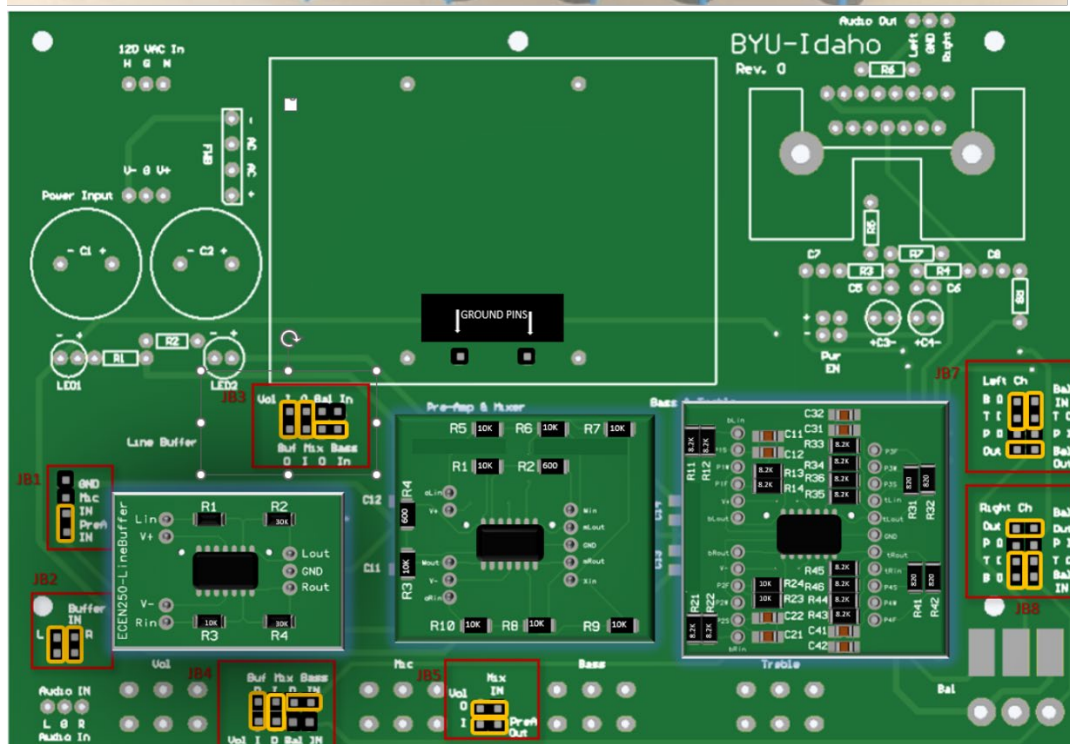
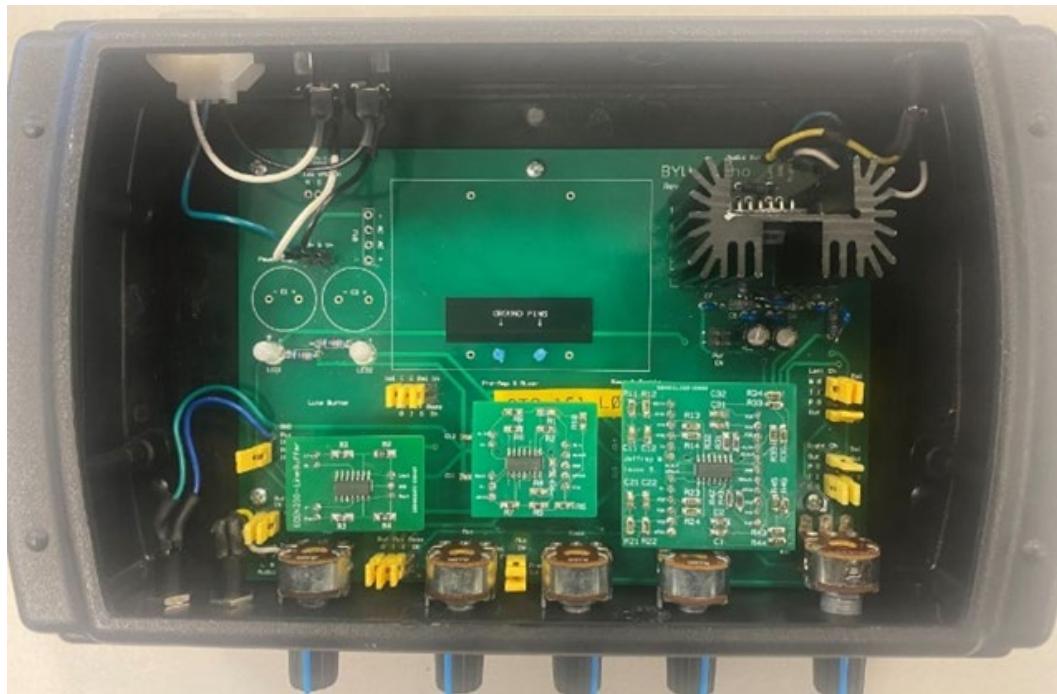


Figure.0 Jumper.Configuration.to.Include.Treble.and.Bass.PCB;

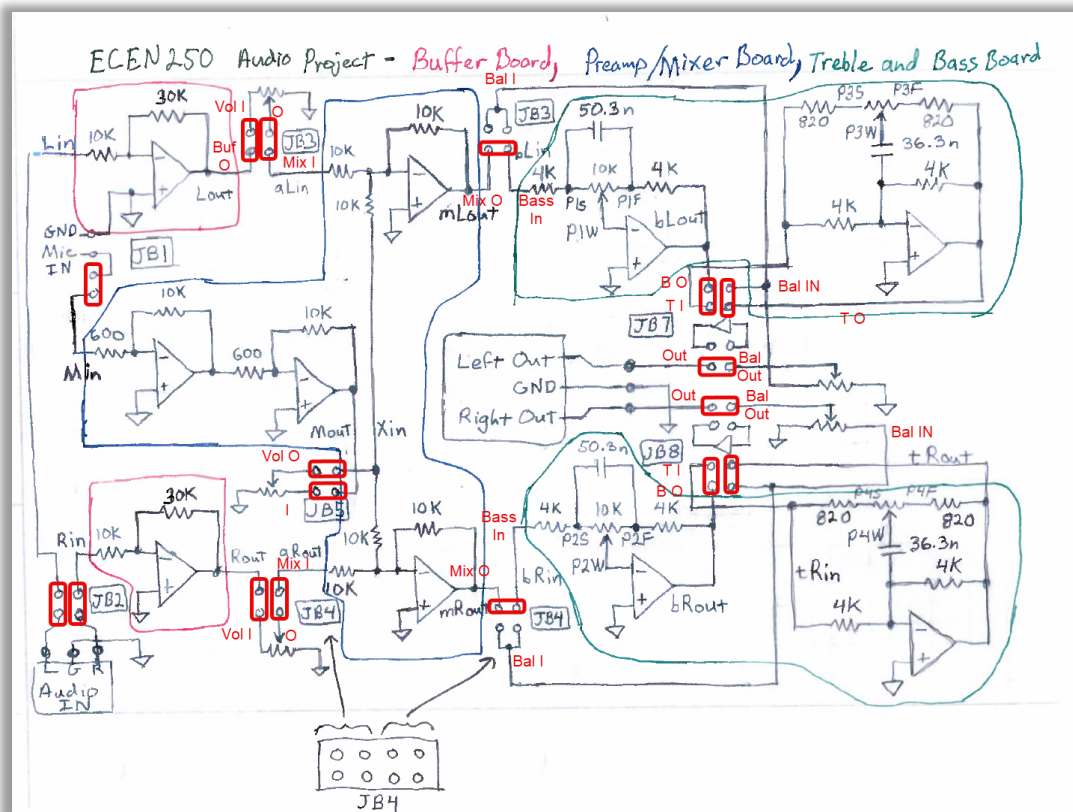


Figure 1 Schematic when all three daughter boards are connected;

Here are some questions to investigate as you determine if the system is working correctly.

- Can you hear low frequencies being attenuated or boosted when the bass knob is turned fully counterclockwise or fully clockwise?
- Can you hear high frequencies being attenuated or boosted when the treble knob is turned fully counterclockwise or fully clockwise?
- Does the microphone volume knob control the volume of the microphone coming out of the speakers?
- Does the microphone come out of both the left and right speakers?
- Does the audio volume knob control the volume of the audio coming out the speakers?
- Does the audio signal come out of both the left and right speakers?
- Do the microphone and the audio signals mix and come out of both the left and right speakers?
- Does the balance knob change the percentage of sound coming out of the left and right speakers?

Pass Off Working System

Have the Teaching Assistant or the instructor pass off your working system.

The appendix is included for your reference.

Appendix A Playing Audio with Only the Buffer PCB

To verify the functionality of the Buffer PCB, we will route an audio signal through the Buffer PCB and then to the audio volume control, and finally to the speakers. Figure 3 shows all the needed jumper connections.

To start, identify the location of the audio jack on the front of the chassis. You can trace its connection to Audio IN (Just below JB2) on the board. Make the jumper connections to connect the audio in Left and Right channels to the Buffer In Left and Right channels respectively as shown in JB2 of Figure 3. To remove the connectors, you can use the needle nose pliers in the drawer.

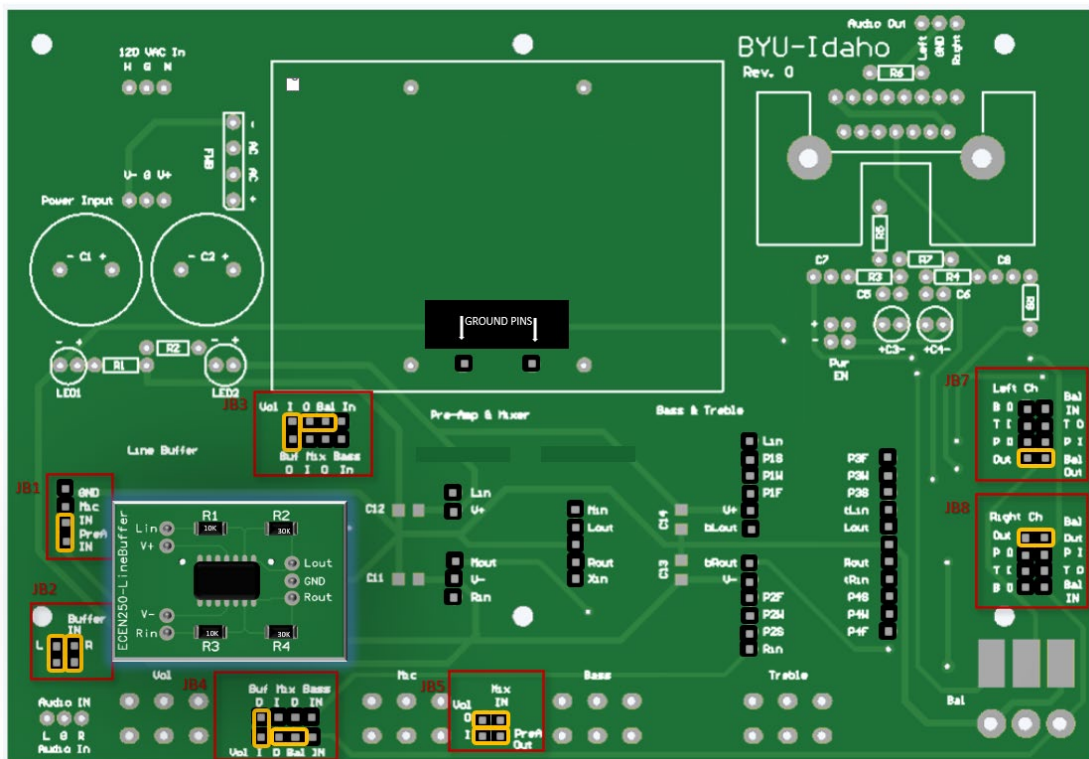


Figure 3: Jumper Connection for playing audio with only the buffer PCB.

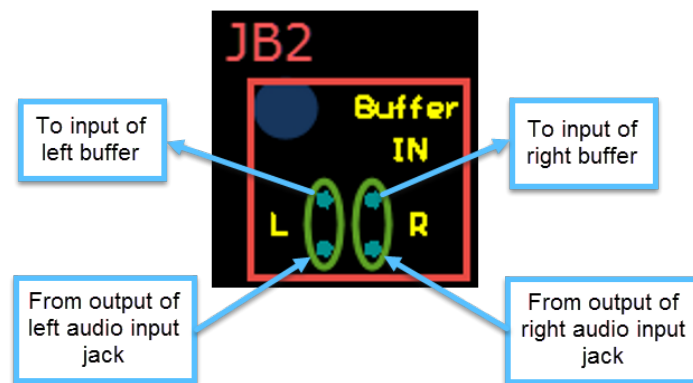
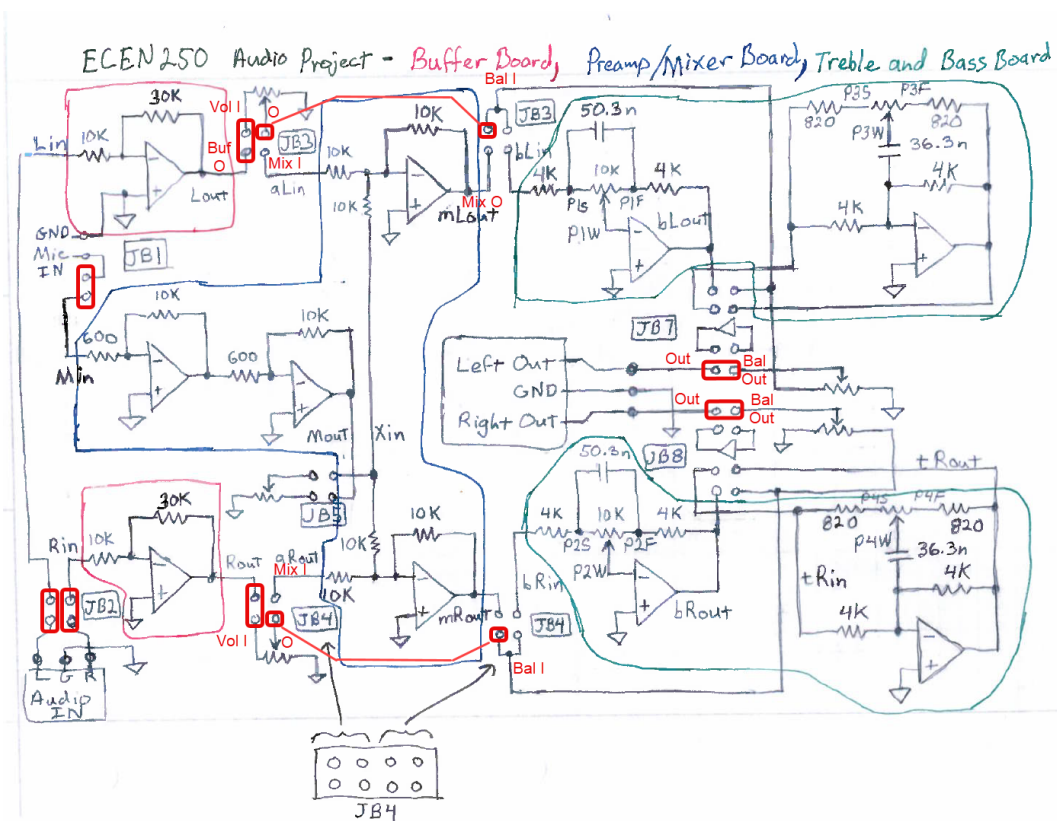


Figure 4 Audio In connections between audio jack and buffer input for both left and right channels explained.

The output of the line buffer (Buf O) connects to region JB3 (Left) and JB4 (Right) as illustrated in Figure 3. The output (Buf O) from the left buffer is connected to the input of the volume control (Vol I). Note, we can make a connection from the output of the volume control to the input of the mixer (Mix I) or bypass the mixer and go straight to balance control. In this section, we bypass the mixer and route the signal straight to the balance control by connecting the output of the audio volume control (Vol O) to the input of the balance control (Bal). Finally, as seen in region JB7 and JB8, the output of the balance

control for both the left and right channels respectively is connected to (Out) which is the input to the speakers.

Note that JB1 and JB5 connections will have no effect on the Buffer PCB because the Preamp and Mixer PCB are not connected, and the signals are routed to bypass this board include the treble and bass boards.

You may now insert your buffer board into the Line Buffer slot of the chassis if you haven't done so already. You can power the board by plugging in the power chord into the chassis and into an outlet. Connect the speakers at the back of the chassis and use the 1/8" stereo audio cable provided to connect to the Audio IN in front of the chassis on one end and into the lab computer on the other end of the cable. Play a song on the lab computer. To play a song, you may browse to this link [Come, Thou Fount of Every Blessing \(2011\) | The Tabernacle Choir – YouTube](#). Verify that you can hear the song playing out of the connected speakers.

Verify Functionality

- What does the audio volume knob do when you turn it? Does it work as expected?
- What does the balance control knob do when you turn it? Does it work as expected?

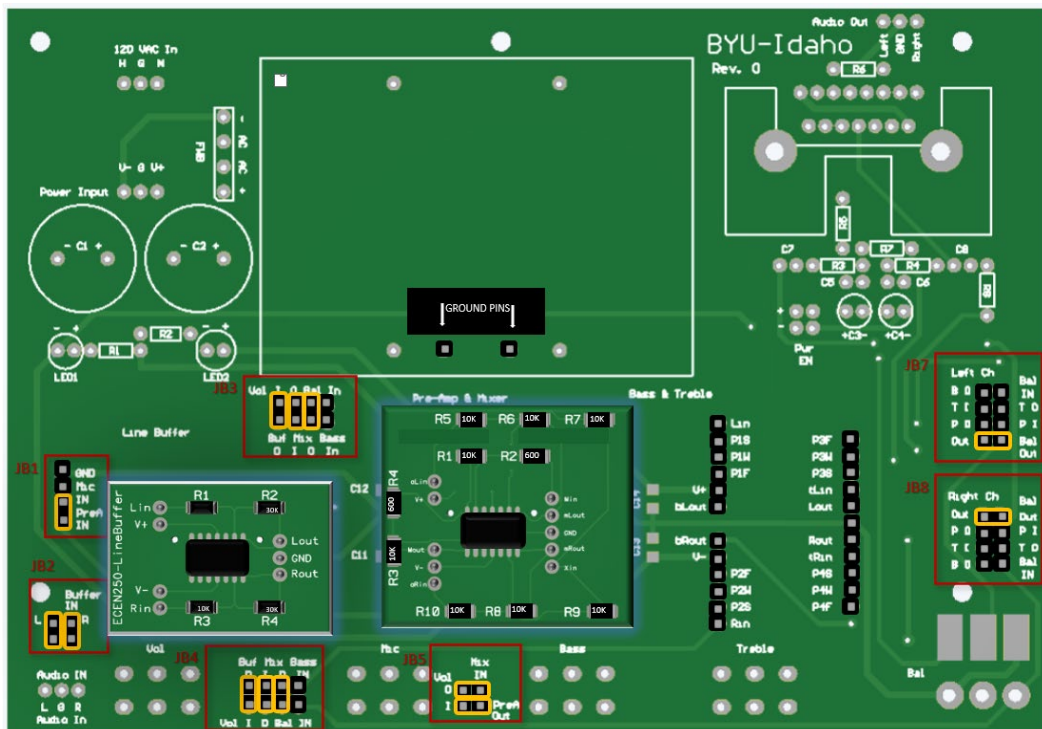
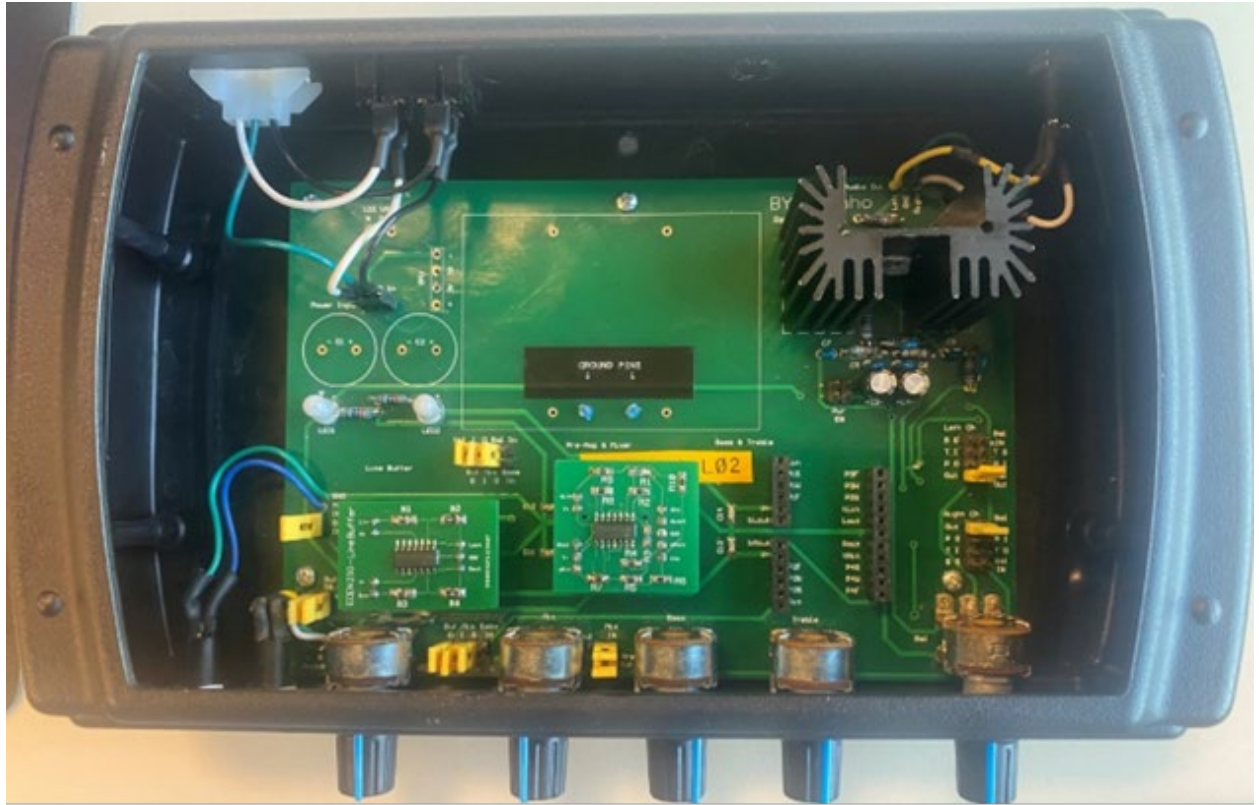
Appendix B Mixing Audio with the Microphone {Buffer PCB and Mic. Preamp and Mixer PCB}

We will now verify the functionality of the Microphone Preamp and Mixing PCB. With the power turned off, connect the header jumpers in accordance with Figure 5. Note that “Mic In” is connected to “PreA IN” (see JB1). The microphone signal is then amplified and the amplified output at “PreA Out” (see JB5) is connected to the Mic ‘Vol I’ which goes to the input to the potentiometer. The jumpers in region JB3 and JB4 are modified to incorporate the Pre-Amp and Mixer into the system.

Connect the pre-amp and mixer PCB into its proper header pins (**make sure the line buffer is also plugged in**). Connect the microphone, powered speakers, and an audio source to the jacks on the chassis. Ensure that the ‘Bal Out’ pins are still connected to the ‘Out’ pins in JB7 and JB8. With the two different volume knobs on the chassis, you should be able to manage the microphone volume separately from the audio volume.

Verify Buffer PCB and Preamp & Mixer PCB Functionality

- Does the microphone volume knob control the volume of the microphone coming out of the speakers?
- Does the microphone come out of both the left and right speakers?
- Does the audio volume knob control the volume of the audio coming out the speakers?
- Does the audio signal come out of both the left and right speakers?
- Do the microphone and the audio signals mix and come out of both the left and right speakers?
- Does the balance knob change the percentage of sound coming out of the left and right speakers?

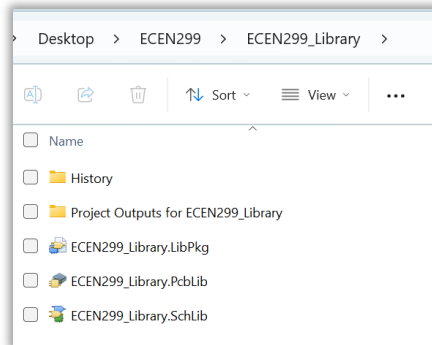


Appendix C – Setup ECEN299 Library (old)

Extract ECEN299 Library Files

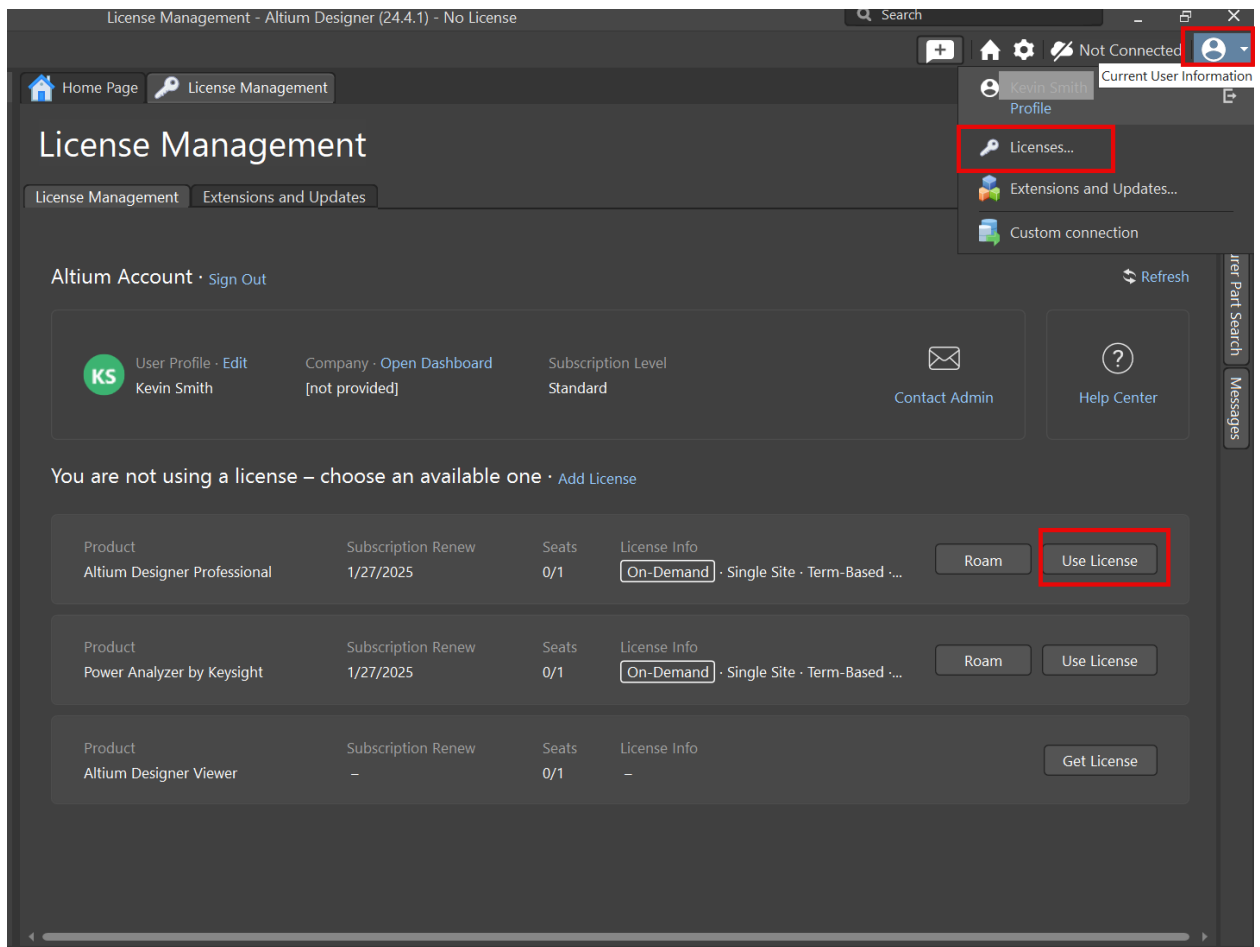
21. Download ECEN299.zip from I-Learn to your Desktop.
22. Extract ECEN299.zip on your Desktop (i.e., %UserProfile%\Desktop).

When complete, the folder ECEN299_Library should be located at the path shown and contain the following files:



Open Altium Designer and Use Your Student License

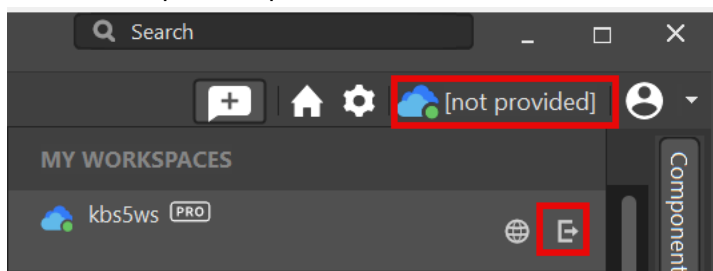
23. Select Windows Open > Altium Designer.
24. Sign In.
25. Select Current User Information > Licenses > Use License.



26. Close Licenses window.

b. Right click on the license management tab and select close license management.

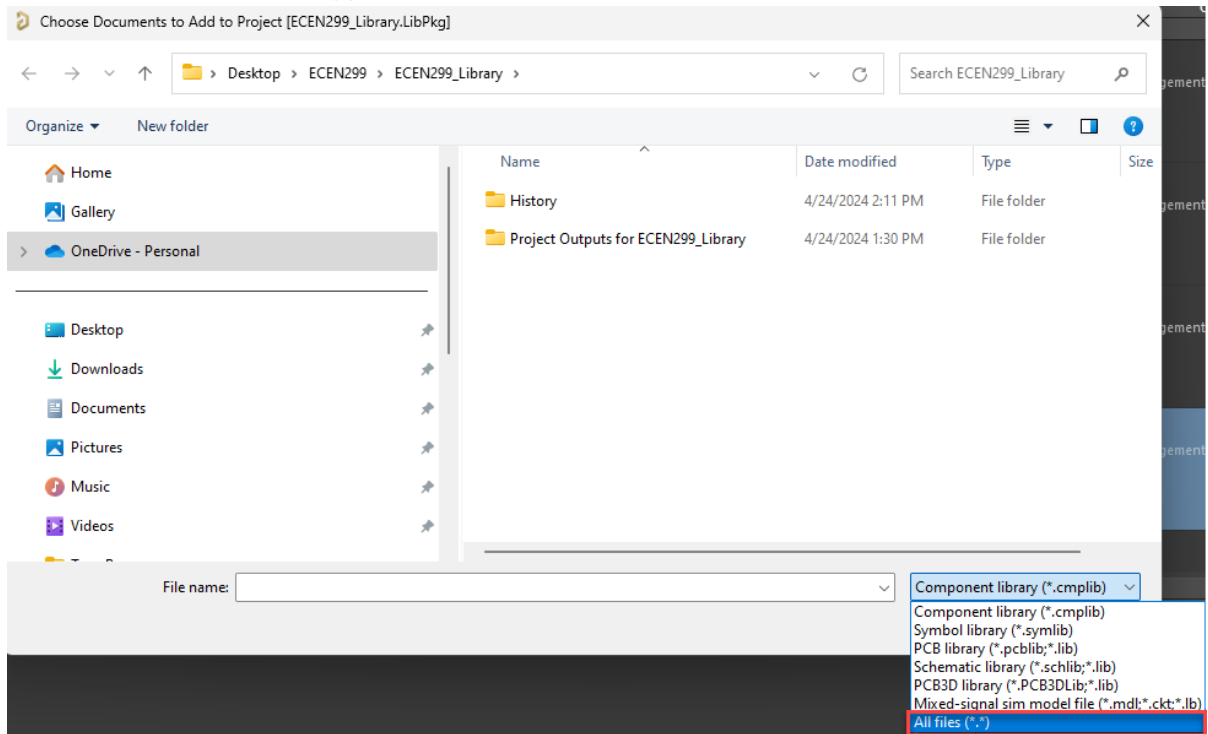
27. Close workspace if opened.



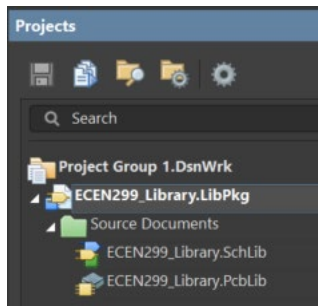
Open ECEN299_Library.LibPkg

1. Select File > Open Project.
2. Select Browse in the lower left corner.
3. Browse to %UserProfile%\Desktop\ECEN299\ECEN299_Library.
4. Select ECEN299_Library.LibPkg.
5. Click Open.
6. If you don't see a list of projects on the left panel, open View > Panels > Select Projects .
7. Right click ECEN299_Library.LibPkg.

8. Select Add Existing to Project.
9. Select All Files as the file type.



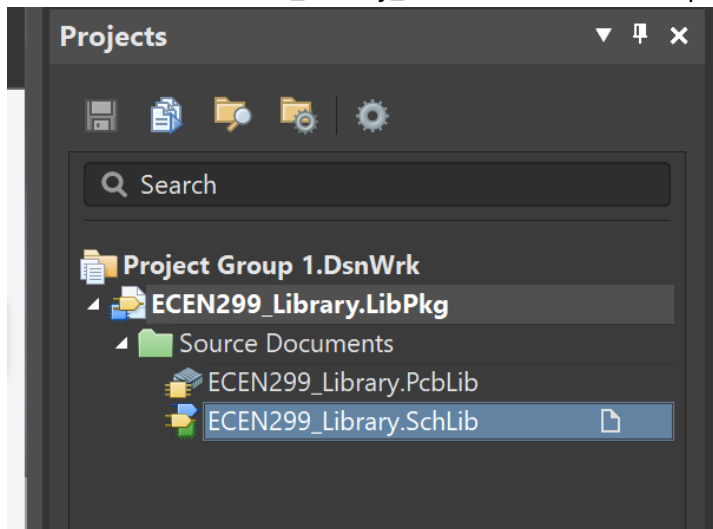
10. Select ECEN299_Library.SchLib and ECEN299_Library.PcbLib.
11. Select Open.
12. Right click ECEN299_Library.LibPkg.
13. Select Save.



View Components in the SCH Library.

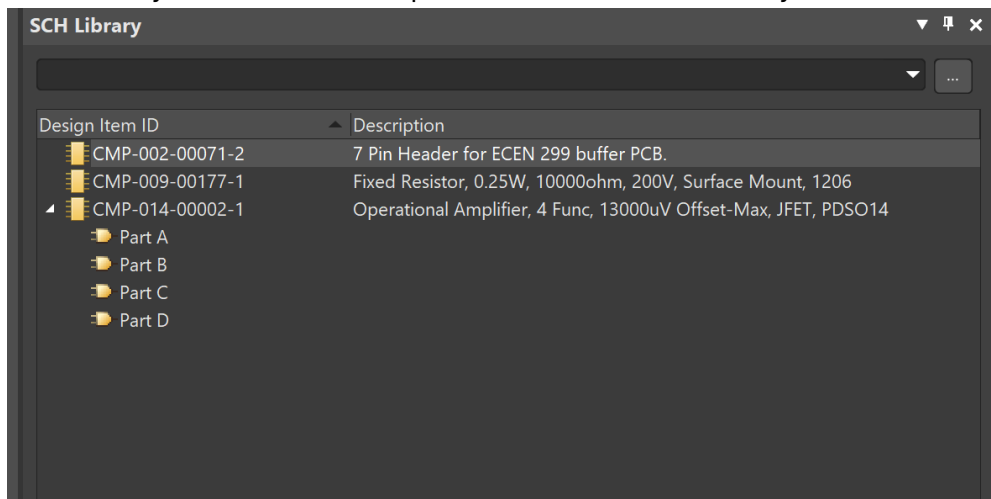
1. If the project panel has closed: Select View > Panels > Projects.

2. Double click ECEN299_Library_Schematic.SchLib to open it.



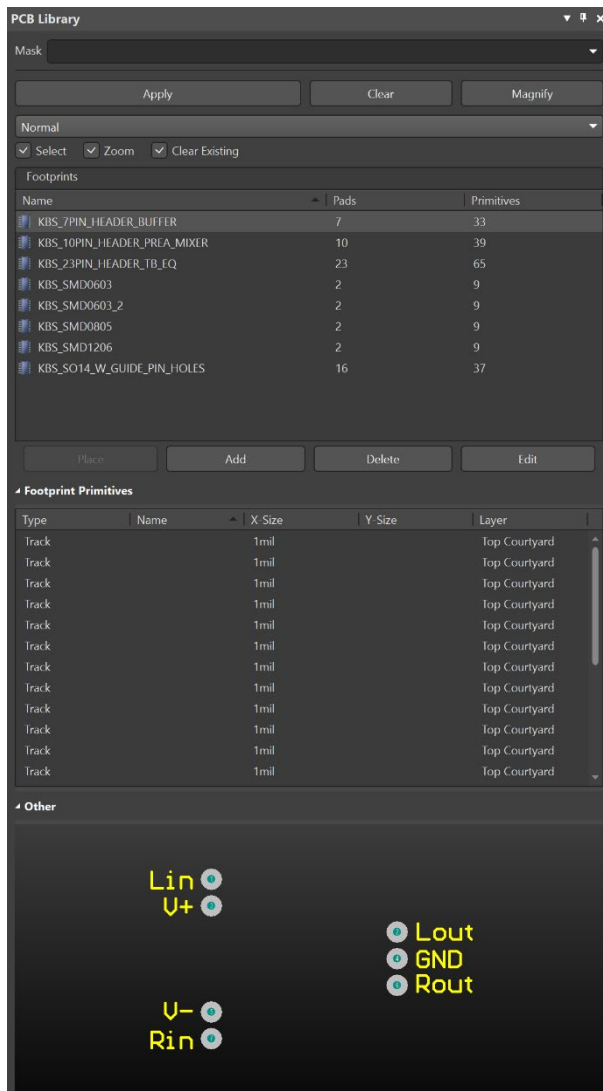
3. Select View > Panels > SCH Library.

This allows you to see what components are in the SCH Library as shown.



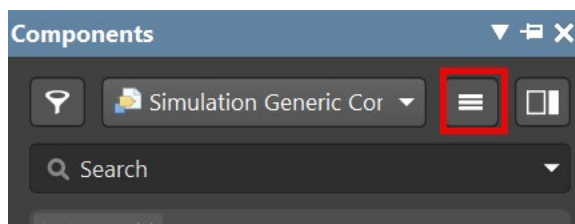
View Footprints in the PCB Library

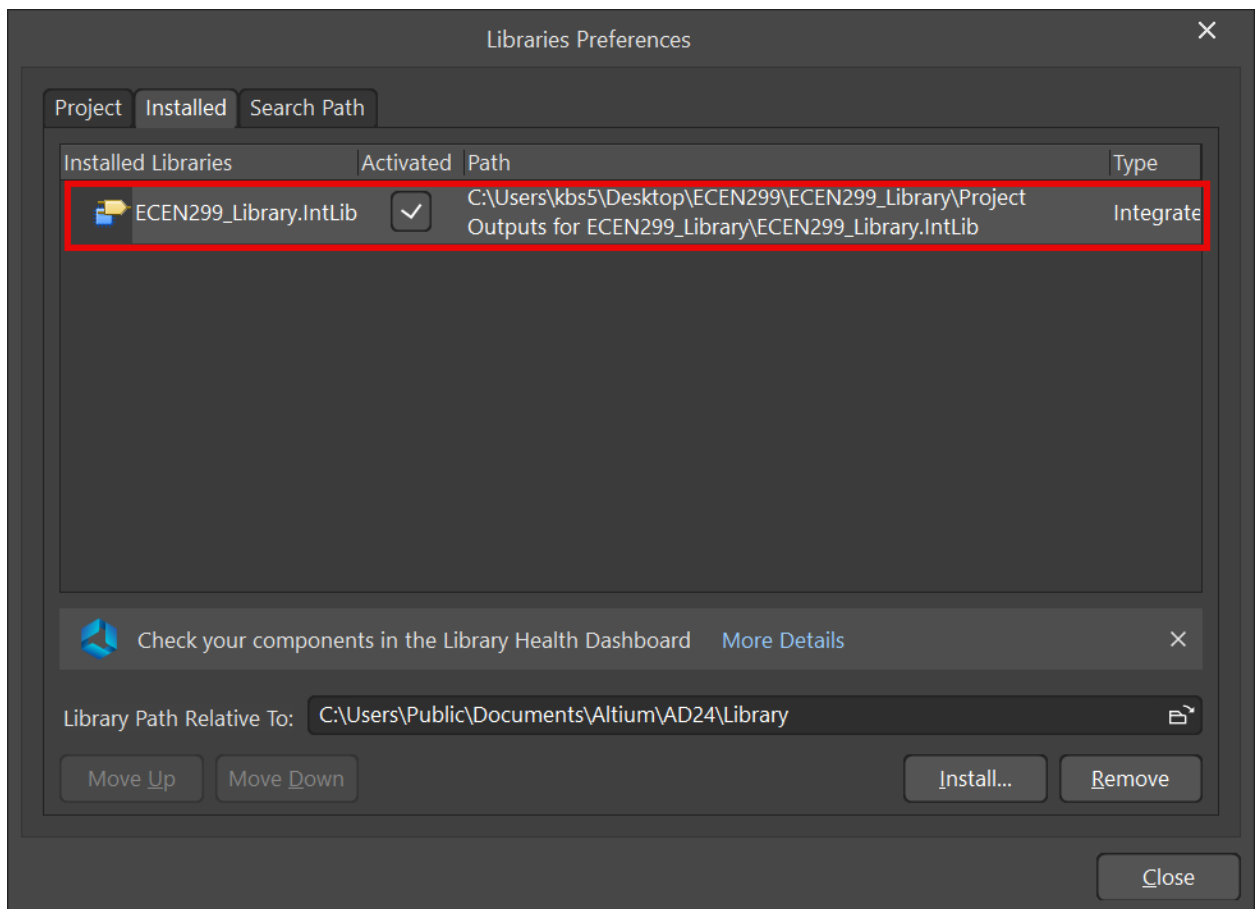
1. Open View > Panels > Projects.
2. Select ECEN299_Library_PCB.PcbLib.
3. Select View > Panels > PCB Library.



Create Integrated Library from the Library Package

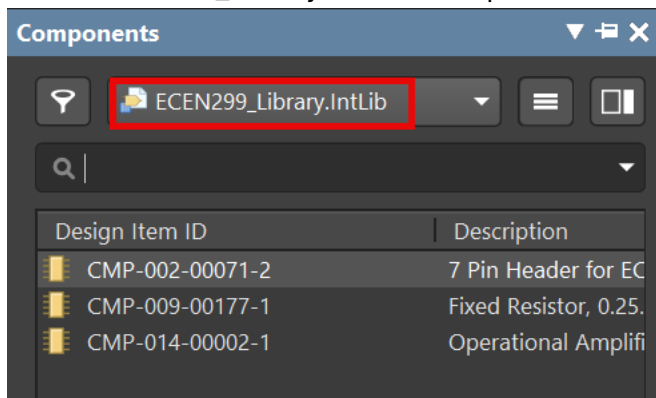
1. Open View > Panels > Projects.
2. Right click ECEN299_Library.LibPkg.
3. Select Compile Integrated Library ECEN299_Library.LibPkg.
This creates and places ECEN299_Library.LibPkg in the subdirectory "Project Outputs for ECEN299_Library" in the folder .\Desktop\ECEN299\ECEN299_Library.
This also installs the library. The path to installed libraries can be seen at View > Panels > Components > triple bar button (Operations) shown below > Libraries Preferences > Installed.





View Components in the ECEN299_Library.IntLib

1. Select View > Panels > **Components**.
2. Select ECEN299_Library.IntLib in the pulldown menu.



The components can be placed on a schematic from here.